Compiler CPCS302

2nd Semester 2021-2022 Mid Exam Female Dep.

This copy is reassembled by students of female department to help other students to practice exam for the subject.

The Mid-term exam covered the ideas of

Ch.1 – Ch.2 – Ch.3

May Allah bless you and good luck.

NO SOLUTION COPY – FOR TRAINING

A solved copy can be found in the Drive too.

King Abdulaziz University

CPCS302 – Midterm Exam

Student Name :....................................

Student ID: ....

Student Section: ....

Grade out of 25	

Q1) Answer the following MCQ by choosing the right answer.

1- What is true about the interpreter among the following choices?
   a- Interpreter is slower than compiler
   b- It converts the source code into machine code
   c- Direct execute the operations specified in the source code
   d- Hard and less efficient in debugging

2- Tokens are the result of the parser, what of the following is incorrect to consider as a token?
   a- Keywords
   b- Identifiers
   c- String of characters
   d- Integers

3- If we have set D = {0,1,2,….,9}, the regular expression that represent writing one D or more is
   a- D
   b- D*
   c- D^+
   d- DD^+

4- If we have a programming language that is working with different platforms, the programming language have
   a- One backend and multi-frontends
   b- Multi-frontends and multi-backends
   c- One frontend and multi-backends
   d- Multi-frontend and one middle end

5- In the concrete syntax tree, the intermediate node is
   a- Operand
   b- Operator
   c- Non termina
   d- Terminal

6- The **r** that is used in the regular expression **r?** produces
   a- Zero or more value of r
   b- One or more value of r
   c- Zero or one value of r
   d- ?

7- The effect of lexical error recovery might create syntax errors if handled by the Scanner
   a- True
   b- False

8- If there was an error of a lexeme, the error shows in the Scanner
   a- True
   b- False

9- The postfix of 8-3+5 is 835-+
   a- True
   b- False

10- If you want to produce a binary pattern of zero's and one's that always ends with four zeros, then the regular expression for this is (0|1)0000
   a- True
   b- False

Q2) Answer the following question

a- Is the following grammar ambiguous or not?

$$E \rightarrow aEE$$
$$E \rightarrow bEE$$
$$E \rightarrow cE$$
$$E \rightarrow \varepsilon$$

1- Ambiguous
2- Unambiguous

b- Derive a parse tree using left derivation to produce this output: *abc*

Q3) This question shows a grammar of some programming language, answer the following questions.

$S \rightarrow tTF \mid \varepsilon$

$T \rightarrow fT \mid \varepsilon$

$F \rightarrow hF \mid \varepsilon$

a- Determine the four tuples of the grammar above

b- Use right derivation to produce this output: *tfh*

Q4) The following grammar is ambiguous, answer the question about this grammar:

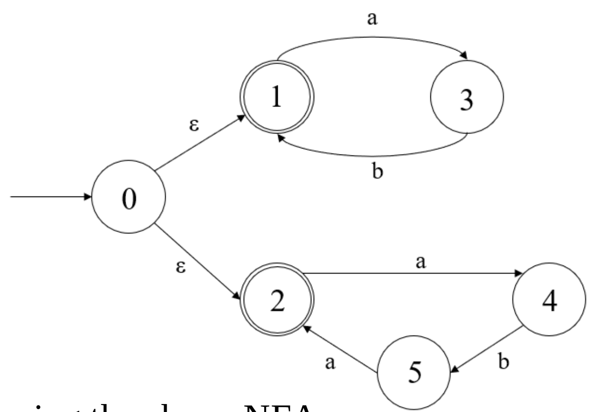
$E \rightarrow E * E \mid E / E \mid D$

$D \rightarrow 0 \mid 1 \mid 2 \mid \ldots.. \mid 9$

a- What is the reason behind the ambiguity?

b- Fix the ambiguity.

Q5) The following graph represent a NFA of a certain regular expression, answer the following question using this NFA.



a- Find the following using the above NFA

F:……………………………………………………………………………….

Σ:……………………………………………………………………………….

b- Fill the following transition table using the above NFA

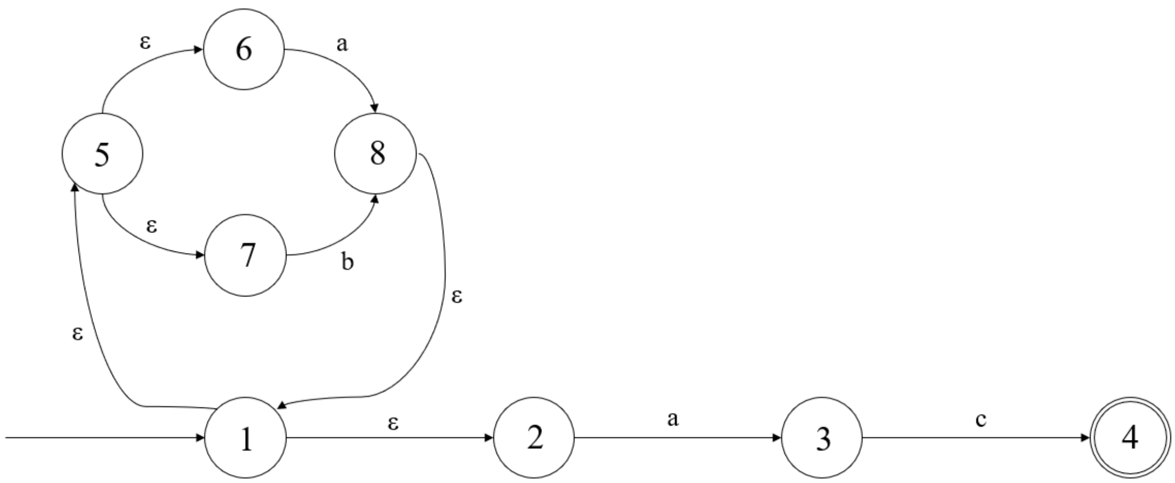
State	a	b	ε
1			
2			
3			
4			
5			

c- Check if the following string and answer the questions.

1- The pattern *abaa* is (Accepted/Rejected) using the above NFA. Draw the path:

2- The pattern *abab* is (Accepted/Rejected) using the above NFA. Draw the path:

Q6) Answer the following questions using this NFA



a- Find the epsilon closure for state {1}:
ε-closure ({1}):

b- Find the transition from the state 1 to the symbol a:
Move (ε-closure ({1}) , a):

c- Find the transition from the state 1 to the symbol b:
Move (ε-closure ({1}) , b):

Q7) Draw the NFA for the following regular expression: 0* | (10)

Note: the (10) is not ten, it's 1 concatenation 0.

Draw solution here: